

Chapter 02

Monte Carlo Method : Monte Carlo is a mathematical or statistical technique by which we can solve a problem with using random sampling, which are otherwise difficult or impossible to solve.

Example (2.1) $\rightarrow$ Area of an Irregular Figure

Enclose the figure in a regular figure of known area, say square of area A . Inside the square figure mark at random, N numbers of points.

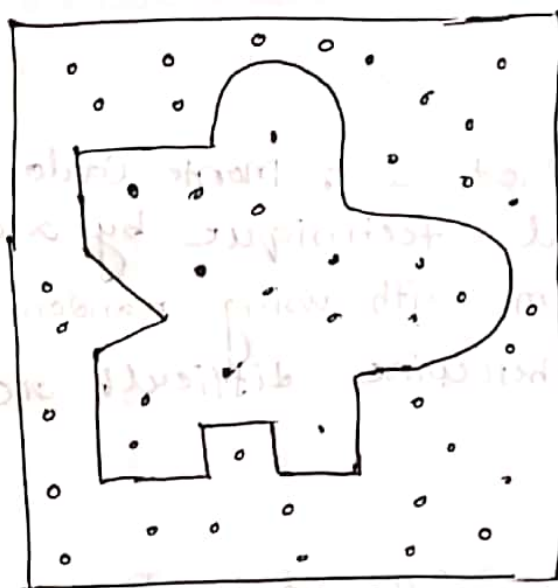
Count the point inside the irregular figure, let these be M , while the points inside the square are N .

Now, it is geometrically obvious that the ratio of area F of the irregular figure to the known area A , would approximately be the same as the ratio M to N .

$$\frac{F}{A} = \frac{M}{N}$$

$$\Rightarrow F = \frac{M}{N} \times A$$

If the number of random points N is increased the accuracy of the result will also increased



Here, $N=84$

$m=26$

$\frac{m}{N} = 0.3095$ while the true ratio is 0.3126

If $A = 9 \times 9 = 81 \text{ cm}^2$, then

$$F = 0.3095 \times 81 = 25.07 \text{ cm}^2$$

For each random point two random numbers will be used to generate the two coordinates of the point. Since our square is $9 \times 9 \text{ cm}^2$. That is each side is 9 cm long. If we take the random numbers between 0 and 1 the the co-ordinates of a random point are obtained by multiplying the random numbers by 9.

SNO	RN_1	RN_2	x_1 $RN_1 \times 9$	y_1 $RN_2 \times 9$
1	.96	.42	8.64	3.78
2	.71	.10	6.39	0.9
3	.03	.30	0.27	2.7
4	.90	.05		
5	.31	.03		

Notes: The error of calculation is proportional to $\frac{1}{N^2}$. Hence the method is suitable for those problems only which do not require a very high degree of accuracy.

Example 02 $\rightarrow$ A Gambling Game

Let us consider a game in which an unbiased coin is repeatedly flipped. For each flip you have to pay Rs 1. and when the difference between heads tossed and tails tossed become 3, you get Rs 8.

If the required difference is obtained in less than 8 flips, you win some money and if it is more than 8 flips, you lose.

When a coin is flipped, the probability of head or tail is same that is 50%, but there are 10 possible one digit random numbers. Thus half of these (0, 1, 2, 3, 4) can be taken for head and the remaining half (5, 6, 7, 8, 9) for the tail.

Let simulate some games in table:

For first 7 games we find that total money paid in Rs $7+9+5+11+11+9+7 = 59$, while the return in Rs 56.

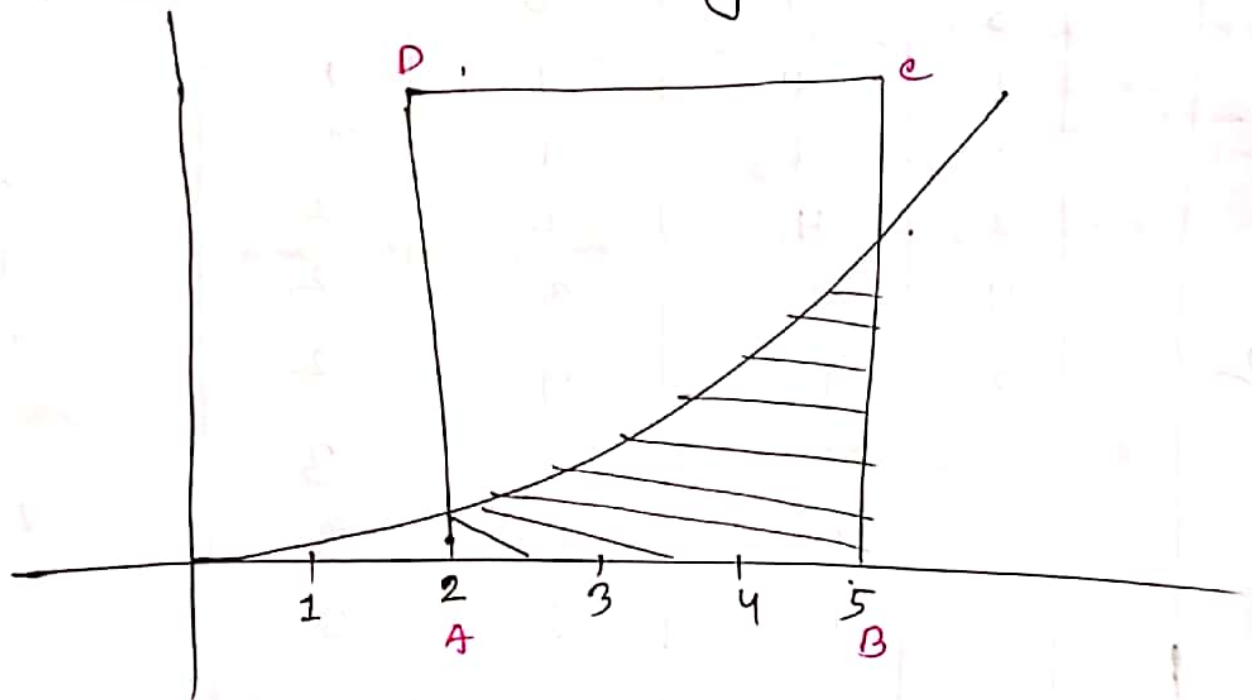
Gr No	SNO	RN	Head/ Tail	Cumulative		Differences	w/ RS
				Heads	Tails		
1	1	5	T	0	1	1	1
	2	9	T	0	2	2	
	3	3	H	1	2	1	
	4	6	T	1	3	2	
	5	4	H	2	3	1	
	6	8	T	2	4	2	
	7	6	T	2	5	3	
2		8	T	0	1	1	Loss Rs 1
		1	H	1	1	0	
		5	T	1	2	1	
		2	H	2	2	0	
		4	H	3	2	1	
		0	H	4	2	2	
		6	T	4	3	1	
		3	H	5	3	2	
		1	H	6	3	3	

Example 2.3 $\rightarrow$ Numerical Integration by Monte Carlo method

Let Consider single variable function

$$I = \int_2^5 x^3 dx$$

$\therefore$ Exact value of the integral $I = \left[\frac{x^4}{4} \right]_2^5 = 152.25$



shaded value shown the value of the integral
let this area be enclosed in a rectangle $ABCD$
whose area is known $(140 \times 3) = 420$.

Let N darts fall in the rectangle $ABCD$ and
out of these let M darts fall in the
shaded area.

Then the ratio of the shaded area I to
the area of rectangle $ABCD$ is M to N .

To mark a random point we need two random numbers to generate the two co-ordinates x and y .

Let us take two digit random numbers between 20 and 50 to represent the x co-ordinate. If r is a random number between 20 and 50 then $x = 0.1(r)$

To generate y co-ordinate take random number between 0 and 1 and multiply it by the height of the rectangle ABCD.

Now check if $y \leq x^3$. If no the point lies below the curve, otherwise it is above the curve.

Simulation procedure is demonstrated in below table. The point which lies below the curve added to m , while N is the number of trials.

SNO	RN	$x = 0.1r$	RN	$y = 140r$	x^3	m	N
22	2.2	0.2	0.57	79.8	10.65	0	1
25	2.5	0.25	0.18	25.2	15.63	0	2
18	1.8	0.18	0.00	0.0	5.83	1	3
45	4.5	0.45	0.90	126.0	91.13	1	4
25	2.5	0.25	0.05	7.00	15.63	2	5
27	2.7	0.27	0.77	107.8	19.68	3	6
48	4.8	0.48	0.66	92.4	110.60	3	7
.	.	.	.	.	.	.	.
.	.	.	.	.	.	.	.
.	.	.	.	.	.	.	.
.	.	.	.	.	.	8	20

Then the value of integral is $I = \frac{8}{20} \times A$
 $= 0.4 \times 420 = 168.00$

where actual value of the integral is 152.25.

By increasing the number of points ~~can be~~
~~generated by a computer~~ ~~obtained~~ ~~and~~ ~~the~~
 value of I quite close to the actual
 value can be obtained, and this large number
 of points can be generated by a computer

Now check if $y = x^2$ the points are below
 the curve or above it in order to find the area

Since the points are below the curve in order to find the area
 to be added to the area of the rectangle

x	y	$2x$	y^2	$4x$	$4y$	$4x^2$
0	0	0	0	0	0	0
1	1	2	1	4	4	4
2	4	4	16	8	16	16
3	9	6	81	12	36	36
4	16	8	256	16	64	64
5	25	10	625	20	100	100
6	36	12	1296	24	144	144
7	49	14	2401	28	196	196
8	64	16	4096	32	256	256
9	81	18	6561	36	324	324
10	100	20	10000	40	400	400

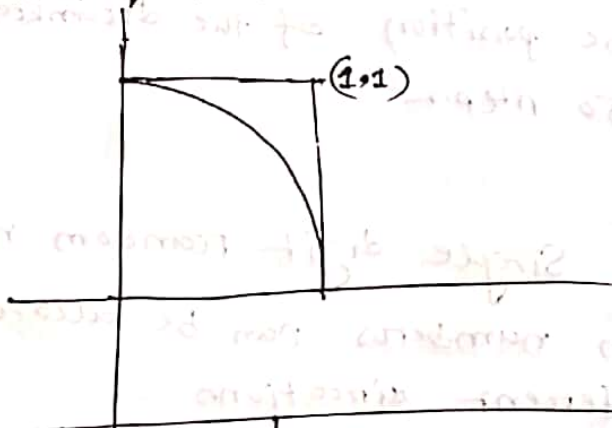
Example 2.4 → Determination of value of π by Monte Carlo Method

The value of π can be determined by using the relation for area of a circle.

$$\text{Area of circle} = \pi r^2$$

$$\text{Area of the quadrant of a unit circle} = \frac{\pi r^2}{4} = \frac{\pi}{4}$$

All points satisfying the equations $x^2 + y^2 \leq 1$; $x, y \geq 0$ lie in this quadrant. $\Rightarrow y \leq \sqrt{1-x^2}$



SNO	x	y	$\sqrt{1-x^2}$	In/out
1	.82	.95	.5724	out
2	.34	.14	.9404	in
3	.20	.84	.9797	in
.	.	.	.	.
.	.	.	.	.
.	.	.	.	.
.	.	.	.	.
.	.	.	.	.
.	.	.	.	.
.	.	.	.	.
.	.	.	.	.
40				

out of 40 points 31 lie within the quadrant

$$\therefore \pi = \frac{31}{40} \times 4 = 3.16$$

Example 2.5 $\rightarrow$ Random walk

A drunkard moves from a point to a destination and takes steps in the forward direction to his left and to his right at random. The length of step is almost constant. The probability of taking a step in the forward direction is 50%, left 30%, right 20%. If the co-ordinate of the starting point are taken as (0,0) and the forward is due y-direction. Find the position of the drunkard after he takes 50 steps.

Soln:

Using single digit random numbers the random numbers can be allocated to steps in different directions.

Direction	probability	RN
F	50%	0,1,2,3,4
L	30%	5,6,7
R	20%	8,9

SNO	RN	Direction	Position	
			x	y
1	6	L	-1	0
2	2	F	-1	1
3	3	F	-1	2
...	1	.		
...	2			
...	4			
...	0			
...	3			
...	2			
...	1			
...	0			
20				

If step is forward then $y++$
 backward " $y--$
 Right $x++$
 Left $x--$

